



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

WEEK	Virtual Lab Commissioning and Network Validation
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Registration No	C11/26/FCDF/17
GUIDED LAB WINDOW	September,2026
SELF-PACED WINDOW	Monday-Friday after the guided session
SUBMISSION DEADLINE	2-October,2026, 23:59 WAT
LMS ITEMS	Lab 1
Linux Version	6.18. 12+kali-amd64

Purpose of this laboratory

This laboratory prepares the complete ICDFA virtual environment for all later exercises. You will reconnect the imported virtual machines to the correct VirtualBox networks, generate unique MAC addresses, confirm the preconfigured IPv4 settings and prove that traffic follows the intended Client, OPNsense, DMZ and VRouter path.

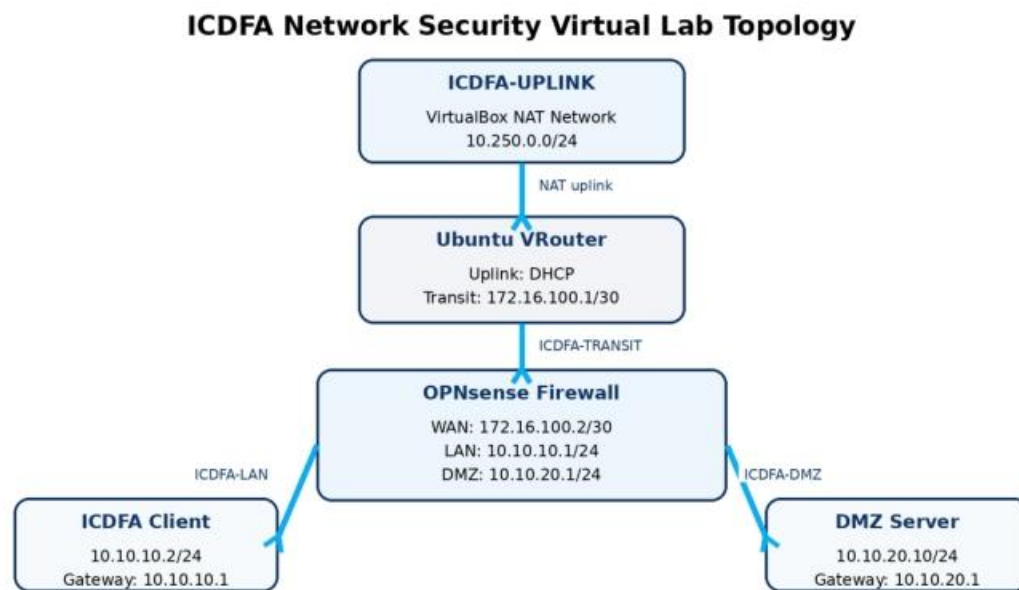
Complete this laboratory before attempting any packet analysis or firewall exercise. At the end, create the required baseline snapshots. Those snapshots give you a safe recovery point if a later configuration change causes a problem.

Learning outcomes

- Create and identify NAT Network and Internal Network segments in VirtualBox.
- Generate and record a unique MAC address for every enabled virtual adapter.
- Map each adapter to the correct ICDFA network segment.
- Verify IPv4 addresses, default gateways, routes, forwarding and required services.
- Complete local, inter-network, DNS, web and upstream connectivity tests.

ICDFA virtual lab topology

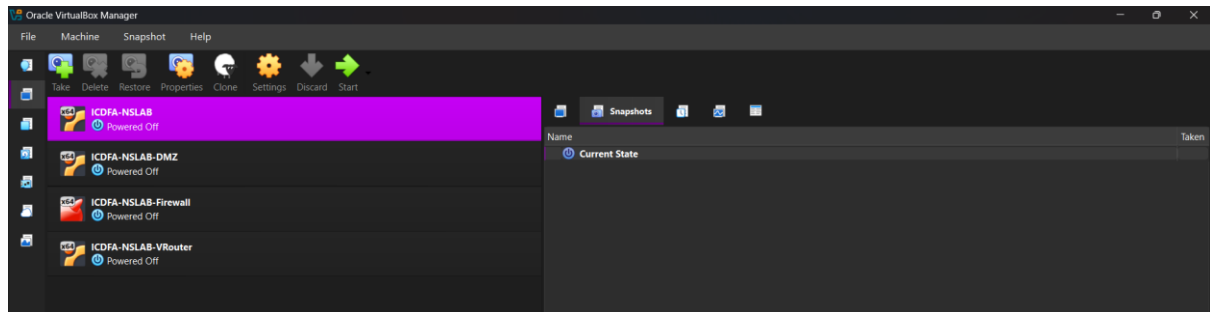
- The lab contains four virtual machines and four logically separate network segments. Traffic between the LAN, DMZ and upstream network must pass through the intended routing and firewall components.



Part 1: Confirm that every VM is powered off

Evidence:

Confirming every VirtualBox Machine is closed.

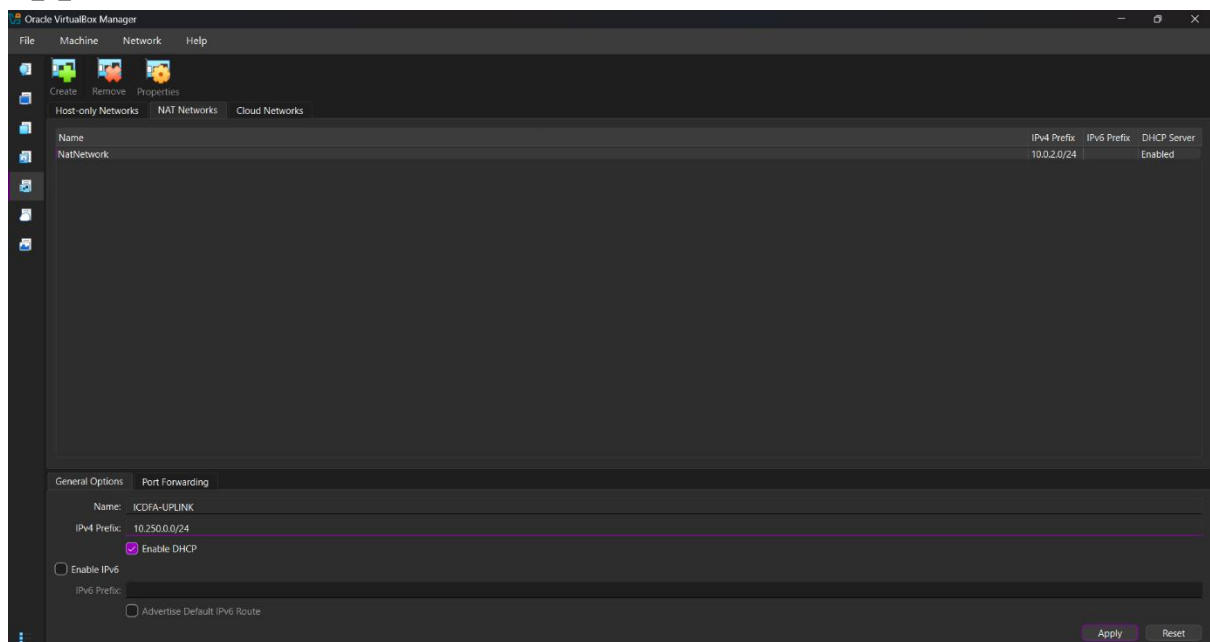


Part 2: Create the ICDFA upstream NAT Network

STEPS:

In VirtualBox Manager, open Tools and then Network. Select the NAT Networks tab. **Then** Create a new NAT Network. Set its name to **ICDFA-UPLINK**. Set the IPv4 prefix to 10.250.0.0/24 and enable DHCP.

Apply the settings and confirm that ICDFA-UPLINK appears in the NAT Networks list.



Part 3 Generate unique MAC addresses and attach the networks

Explanation:

The training appliances were distributed without reusable MAC identities. So, we have to generate new MAC addresses after import. A MAC address must be unique within the lab; duplicated addresses can cause unstable ARP entries and misdirected Ethernet frames.

TABLE:

VM	Adapter	Attachment	Network name	New MAC address
Client	1	Internal Network	ICDFA-LAN	08002735D07A
OPNsense	1	Internal Network	ICDFA-TRANSIT	0800270E86E8
OPNsense	2	Internal Network	ICDFA-LAN	080027952A7D
OPNsense	3	Internal Network	ICDFA-DMZ	0800277E35C1
DMZ Server	1	Internal Network	ICDFA-DMZ	08002716FC38
VRouter	1	NAT Network	ICDFA-UPLINK	08002786C518
VRouter	2	Internal Network	ICDFA-TRANSIT	08002759986F

Part 4:Start the machines in the required order

Explanation:

I have imported all the machines inn Ubuntu and all started them al are in running state.

Part 5: Validate the Client

Evidence:

```
J+l icdra@icdra-ns-lab-client-v1:
icdfa@icdfa-ns-lab-client-v1:~$ ip -4 -br address
lo                UNKNOWN      127.0.0.1/8
enp0s3            UP          10.0.2.15/24
icdfa@icdfa-ns-lab-client-v1:~$
```

PART 6: Validate the OPNsense interfaces

the OPNsense WAN uses a private training address, open Interfaces, WAN and ensure Block private networks is not selected. Save and apply only if you changed the setting.

In System, Gateways, Configuration, verify that VROUTER_GW uses WAN, IPv4 and gateway address 172.16.100.1. The routing status must contain a default route through 172.16.100.1.

EVIDENCE:

```
| Code:      https://github.com/opnsense |
| Reddit:    https://reddit.com/r/opnsense |
-----
*** icdfa-ns-lab-firewall-v1.icdfa.test: OPNsense 26.7 (amd64) ***

DMZ (em2)      -> v4: 10.10.20.1/24
LAN (em1)      -> v4: 10.10.10.1/24
WAN (em0)      -> v4: 172.16.100.2/30

HTTPS: SHA256 E8 0B 30 3E D5 89 EB 63 A0 D8 83 2F E6 38 4C 22
          82 2C 5A A4 39 D5 45 54 57 50 A0 73 58 E0 46 90
SSH:   SHA256 K10LtdmW484dUmuPy37tD714RA0De5LKjSU/+bKB1Tw (ECDSA)
SSH:   SHA256 5wvyP8AD885sP0fHz7CgfNSTfMhPwXXX42x23cBxco4 (ED25519)
SSH:   SHA256 6e/0Xr jsdfL/Hsy8atZw0IPQGsb1P6zFfu j/FZ50d +A (RSA)
```

CONNECTIVITY TEST:

```
icdfa@icdfa-nslab-client-v1:~$ ping -c 3 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=27.2 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=28.5 ms

--- 10.10.10.1 ping statistics ---
3 packets transmitted, 2 received, 33.3333% packet loss, time 2019ms
rtt min/avg/max/mdev = 27.228/27.863/28.499/0.635 ms
icdfa@icdfa-nslab-client-v1:~$ ping -c 3 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=64 time=25.6 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=64 time=23.8 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=64 time=23.6 ms

--- 1.1.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 23.598/24.332/25.601/0.900 ms
icdfa@icdfa-nslab-client-v1:~$
```

CONCLUSION:

The IC DFA virtual network security lab was successfully deployed, configured, and validated. All four virtual machines (Client, OPNsense Firewall, DMZ Server, and VRouter) were properly mapped to their assigned internal networks (IC DFA-LAN, IC DFA-DMZ, IC DFA-TRANSIT) and NAT uplink (IC DFA-UPLINK) with unique MAC addresses to prevent ARP conflicts. IP addressing, routing tables, and connectivity across all segments were verified. Finally, LAB-BASELINE-VALIDATED snapshots were created across all VMs to serve as a clean recovery point for future network security labs.